Gamepad 1: DRIVER

Drives the robot. Pick it on the Driver Station with START + A. Same in every TeleOp (DriverControls.java).

Left trigger (LT / L2)
TURBO (hold)
full speed: 100%

Left bumper (LB / L1)
SLOW (hold)
lining up: 40%

Dpad UP
(free)

Dpad LEFT
(free)

Dpad RIGHT
(free)

Dpad DOWN
(free)

Left stick
DRIVE
almost straight = exactly straight (±15°)

Left stick: press it down
(free)

Right trigger (RT / R2)
TURBO (hold)
full speed: 100%

Right bumper (RB / R1)
SLOW (hold)
lining up: 40%

Y (triangle)
(free)

X (square)
(free)

B (circle)
(free)

A (cross)
STRAIGHT ASSIST on/off
rumble: 1 blip = ON, 2 = OFF. Starts ON

Right stick
TURN
push left / right

Right stick: press it down
(free)

Three driving speeds

SLOW hold either bumper: 40% power, for lining up
NORMAL nothing held: 70% power, everyday driving
TURBO hold either trigger: 100% power, crossing the field
Both held? SLOW wins (the safer choice).
The Driver Station shows "Drive speed" during the match.
Speeds are a first try: tune SLOW_MODE_SCALE, NORMAL_SCALE and TURBO_SCALE in DriverControls.java with the drivers.

Good to know

Directions match the robot: stick forward = drive forward, stick left = strafe left, right stick left = turn left.
Straight-line assist (A): left stick within 15° of straight forward / back / left / right drives exactly straight (#67).
Let go of the sticks and the robot coasts to a stop (FLOAT, decision 023).
Nothing moves during INIT (game rule G304.H).
Free buttons (B X Y, dpad, stick presses) can get driver jobs later, e.g. "turret to forward".

Gamepad 2: OPERATOR

Runs the mechanisms. Pick it on the Driver Station with START + B.

Left trigger (LT / L2)
EJECT (hold)
water wheel DOWN + intake OUT; then all off

Left bumper (LB / L1)
(free)

Dpad UP
(free)

Dpad LEFT
(free)

Dpad RIGHT
(free)

Dpad DOWN
(free)

Left stick
TURRET (left / right)
further = faster; let go = stop. Slow until homed

Left stick: press it down
(free)

Right trigger (RT / R2)
(free)

Right bumper (RB / R1)
(free)

Y (triangle)
LAUNCH (hold)
trigger moves out of the way; let go = blocked

X (square)
WATER WHEEL UP (toggle)
press = on, press again = off

B (circle)
WATER WHEEL DOWN (toggle)
press = on, again = off. X and B cancel

A (cross)
INTAKE (toggle)
press = on, press again = off

Right stick
(free)

Right stick: press it down
(free)

How it works

Flywheel: starts when START is pressed and runs the whole match.
It is NOT on a button. Watch "Launcher Status" for READY.
Toggles (A, X, B): press once = on, press again = off.
X and B cancel each other: press B while going UP = going DOWN.
EJECT beats everything; let go and the intake + water wheel are OFF.
Lights: PURPLE = the next ball to launch is a NECTAR.

Not on a button yet

Turret to forward, hood up / down, turret auto-aim (later, #7). Flower flipper: waiting to see if it's built.
Free: dpad, bumpers, right trigger, right stick.
Code: opmodes/teleop/OperatorControls.java